

TryNex Lifestyle

Full Technical Documentation

Premium Custom Apparel · Bangladesh

Platform:	Full-stack e-commerce + Design Studio
Market:	Bangladesh (all 64 districts)
Domain:	trynexshop.com
API:	trynex-api.onrender.com
Stack:	React 19 + Vite 7 + Express 5 + PostgreSQL
Generated:	31 May 2026

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§1 Platform Overview & Architecture

Dev Environment	Replit (NixOS)	Replit	Both services run side-by-side, pnpm ws
Cache	Upstash Redis (REST)	Upstash	Settings, product list, rate limit counters
Full-stack e-commerce platform			Product images, original design files
			Monorepo Layout
			• artifacts/trynex-storefront/ — React 19 + Vite
			• artifacts/api-server/ — Express 5 REST A
			• artifacts/api-worker/ — Cloudflare Worker
			• artifacts/mockup-sandbox/ — Isolated Vite c
			• lib/db/ — Drizzle schema, 12 mi
			• lib/api-spec/ — OpenAPI 3 spec + g
			• lib/api-zod/ — Generated Zod valida
			• lib/api-client-react/ — TanStack Query ho
			• scripts/ — Seed, PDF generation
			• .github/workflows/ — deploy.yml (main

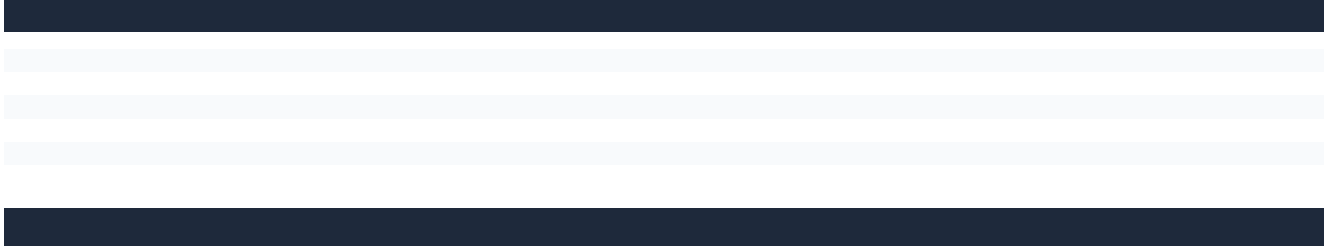
@upstash/redis
DOMPurify

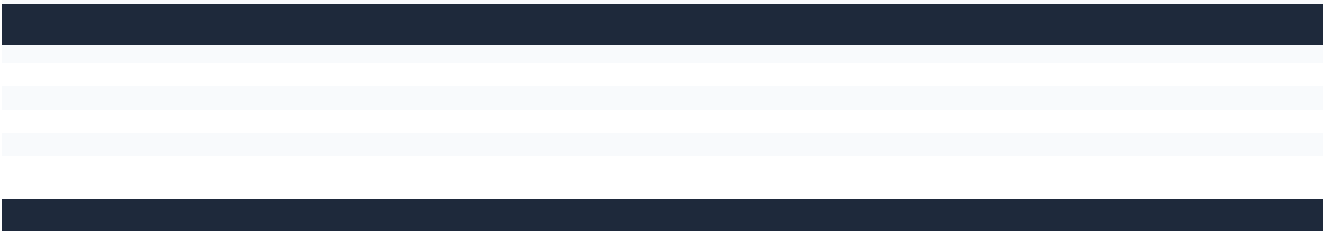
latest
latest
latest

- In-app Redis REST client
- Sanitise user-generated HTML (blog content)
- Non-blocking toast notifications
- Client-side order summary / PDF generation

- Cloudflare R2 — object storage for product images + design originals
- Cloudflare Pages — static SPA hosting, global CDN, custom domain t
- Render.com — Node 24 web service for Express API (auto-deploy on
- Upstash Redis — REST-based Redis for caching and rate-limit counte
- Neon PostgreSQL — serverless PG fallback chain (DATABASE_URL_
_FAILOVER)
- GitHub Actions — 5-job CI/CD: TypeCheck ! CF Pages ! CF Worker !

[illegible]





§5 Database Schema (19 Tables)

[illegible]

§6 Authentication System

Customer Auth

- Registration: email + password (bcrypt 10 rounds) + phone (required for COD)
- Login returns: short-lived JWT access token (1h) + HttpOnly refresh cookie (30d)
- Refresh: POST /api/auth/refresh reads HttpOnly cookie, issues new access token
- Google OAuth 2.0 via /api/auth/google callback (GOOGLE_CLIENT_ID required)
- Facebook OAuth via /api/auth/facebook callback (FACEBOOK_APP_ID required)
- Guest checkout: order placed without account !' auto-creates customer record !' emails credentials
- Password reset: email token (1h TTL) !' POST /api/auth/reset-password
- JWT_SECRET must be set; defaults to random on dev (stateless, not revocable)

Admin Auth

- Single admin account seeded via ADMIN_PASSWORD at startup (bcrypt 10 rounds)
- Login returns: 30-min admin JWT signed with ADMIN_JWT_SECRET (separate from customer secret)
- Sessions stored in admin_sessions table — each token hash is persisted and revocable
- Optional TOTP 2FA: enable in admin !' generates otp1ib TOTP secret !' scan QR in authenticator app
- All admin routes guarded by verifyAdminToken middleware (validates JWT + checks not-revoked in DB)
- Admin JWT secret must be 32+ chars in production (hard exit at boot if too short)
- Admin sessions expire after 30min of inactivity (lastUsedAt rolling window)
- IP address logged per session for audit purposes

Security Headers

- helmet() sets: X-Content-Type-Options, X-Frame-Options (SAMEORIGIN), X-XSS-Protection, HSTS (prod)
- CORS: strict ALLOWED_ORIGINS allowlist; production refuses to start if ALLOWED_ORIGINS is unset
- Content-Security-Policy: configured via helmet; frames blocked except SAMEORIGIN
- No credentials are ever logged; Pino logging masks passwords/tokens by field name

§7 Design Studio — Technical Deep Dive

The Design Studio (/design-studio) is the most complex feature. It renders a 1000×1000 SVG canvas with real-time photographic garment mockups, a drag-and-drop layer system, AI-powered tools, and a 3D preview. It is self-contained in `artifacts/trynex-storefront/src/pages/DesignStudio.tsx` (~4300 lines).

Canvas & Coordinate System

- All products share a unified 1000×1000 SVG viewBox
- PrintZone {x, y, w, h} defines the printable area per product face in 1000-unit space
- Layer positions are stored in the same 1000-unit space — compositing is coordinate-space-invariant
- Canvas renders inside a <svg> element; layers are absolutely-positioned children
- Pinch-to-zoom and pan are implemented via pointer events on the SVG container
- Undo/redo stack: full layer state snapshot on every mutating action (30-deep ring buffer)

Photographic Mockup System (GarmentSVG)

- All garments use real studio photography (transparent PNG, 1000×1000 px)
- Colour rendering: `isNearBlack(hex) = luminance < 0.12` !' use dedicated black photo; all other colours use white base photo + SVG multiply-tint filter
- SVG tint filter pipeline: `desaturate !' feFlood(tintColour) !' feComposite(IN sourceAlpha) !' feBlend(multiply, grayscale) !' feComposite(IN sourceAlpha) = realistic fabric colour`
- `isLightTint(hex) = luminance > 0.92` !' no tint applied (white garment shows as-is)
- Curvature vignette: radial gradient darkens edges of garment photo to simulate 3D fabric wrap
- `600h:400w:1000h:1000w` `waterbottle` `x:348 y:278 w:290 h:575` `waterbottle`
- `600h:400w:1000h:1000w` `highlight` `x:342 y:305 w:316 h:248` `cap`

Layer System

- Layers: each has id, type (neck-label)
- Image layers: src (R2 URL)
- Text layers: text, fontFamily
- Multi-select: Shift+click; g
- Layer lock: locked layers
- Copy front! back helper:
- Design drafts: auto-save

AI & Enhancement Tools

- AI Art generation: POST
 - Background removal: POST
 - HD Upscale: POST /api/
 - Custom colour picker: ST
- only)
- Template presets: 6 built
 - Fonts: 20 typefaces inclu

§8 3D Viewer & Composer

React Three Fiber 3D Preview

- 3D view toggled via the 2D/3D switch in the Design Studio header
- T-Shirt + Long Sleeve: real GLB mesh (models/tshirt.glb, models/longsleeve.glb) with UV-mapped canvas texture
- Hoodie: GLB mesh (models/hoodie.glb) with separate body + hood materials
- Cap: GLB mesh (models/cap.glb) with front panel UV mapping
- Mug: procedural Three.js CylinderGeometry with wrap texture (360° repeat)
- Water Bottle: procedural TubeGeometry with carabiner assembly (cap + body + clip geometries)
- OrbitControls: drag to rotate, scroll to zoom, auto-damping for smoothness
- Garment colour tinting: ShaderMaterial with uniform for colour value; multiply blend in shader
- Canvas texture: composed by composer.ts at up to 2048×2048px for crisp print quality

composer.ts — Canvas Texture Composition

- `composeGarmentMockup(layers, printZone, baseHeight, options) !' HTMLCanvasElement`
- Renders each layer (image/text) onto a 2D canvas in printZone coordinate space
- destination-in alpha-mask clips the composed artwork to the print zone outline
- Supports front + back face composition for multi-face products
- Text rendering: uses same font stack as SVG preview for WYSIWYG consistency
- Output fed to Three.js as CanvasTexture (auto-updates when layers change)
- Same composer used for: 3D preview, 2048×2048 order thumbnail (sent to admin), cart preview card

\$9 Order & Payment Flow

cancelled
delivered

Cancelled (before shipping)
Confirmed, received by customer
Out for delivery

delivered

Payment Methods

- Cash on Delivery (COD): 15% advance required !' customer provides transaction ID
- bKash, Nagad, Rocket, Upay: manual MFS (mobile financial service)
- Card: manual card payment link or in-person (offline)
- Payment status flow: pending !' submitted (customer pays)
- Admin verifies by checking MFS app logs and updating status

Shipping Calculation (Server-Side)

- Free shipping threshold: ₹3,000 by default (configurable)
- Base shipping fee: ₹300 + 10% of order value (capped at ₹1000)
- Promo codes can grant freeShipping flag (overrides threshold)
- Both fees are computed server-side — client subtotal + shipping = total
- District is captured but not used for tiered pricing (flat rate)
- Hamper items: server re-computes price from real price

Custom Design Order Flow

- User designs in Design Studio !' 2048px canvas thumbnail generated
- On 'Add to Cart': thumbnail uploaded to R2 as thumb_nano.jpg, metadata saved to design.json
- Cart stores R2 thumbnail URL + presigned design URL
- On order placement: R2 URLs included in order item metadata
- Admin can download original design file via GET /api/design/{id}/original (presigned URL)
- studioAssetsMissing flag set if R2 upload fails during cart creation

§10 Admin Panel — Full Feature List

Dashboard (AdminOrders)

- Auto-refreshes every 3 seconds via React Query polling
- BroadcastChannel cross-tab sync: order status changes propagate to other open admin tabs instantly
- Order cards show: TN-XXXXX number, customer name, phone, address, total, items summary, payment status badge
- Quick status workflow buttons: Confirm !' Ship !' Mark Delivered
- Quick payment verification: Verify button sets paymentStatus=verified
- Bulk actions: bulk mark as shipped, bulk cancel
- Export: download orders as CSV (date range, status filter)
- Design download: per-order 'Download Design' button !' generates fresh presigned R2 URL
- Order messages: in-line chat thread per order for customer communication

Product & Content Management

- Products: CRUD, multi-image upload (drag-to-reorder), R2 storage, color variants (name+hex+image per variant)
- Categories: create/rename/delete (reassign products before delete), category image
- Blog: full rich-text editor, SEO meta fields, tag management, AI content assist, schedule publish
- Testimonials: create/approve/delete; shown on homepage
- Reviews: approve/delete customer product reviews
- Hero banners: image + link + CTA text, drag to reorder, enable/disable per banner
- Hampers: build gift hamper packages with product picker, pricing, packaging images
- Promo codes: flat/percent discount, min order value, max uses, expiry, virtual (with specific items)

Settings & Configuration

- Studio T-Shirt Price, Studio Mug Price — controls checkout price for Design Studio orders
- Studio T-Shirt Colors, Studio Mug Colors — comma-separated hex list for Design Studio colour swatches
- Free Shipping Threshold, Shipping Cost — overrides defaults (Ÿ3 S Â 6100)
- Site Name, SEO Description, OG Image — global SEO meta defaults
- Google Analytics ID, Meta Pixel ID — injected into storefront HTML at runtime
- Announcement Bar messages — scrolling ticker text
- WhatsApp number — used in the floating WhatsApp button
- Social links (Facebook, Instagram, YouTube, TikTok)
- TOTP 2FA toggle for admin account
- AI Assistant: uses OpenAI-compatible API (configurable endpoint + model)

			429
Storage upload	15 min	40	429
Remove-hq / AI tools	15 min	15	429

Additional Security Controls

- DOMPurify: all user-generated HTML (blog content) sa
- Presigned URLs: 15-minute TTL for private design file
- Admin JWT: separate signing secret, stored hashed in
- ALLOWED_ORIGINS: production server hard-exits on
- Admin JWT secret: server hard-exits on startup if < 32
- Storage backend guard: server hard-exits if storage re
- bcrypt: 10 rounds for all passwords; timing-safe compa
- SQL injection: Drizzle ORM uses parameterised queri
- CSRF: JWT in Authorization header (not cookie) for cu
- Input validation: Zod schemas on all POST/PUT endpo

§12 SEO, Structured Data & Performance

Structured Data (JSON-LD)

- Organization — brand name, logo, sameAs links (FB, IG, YouTube)
- WebSite + SearchAction — enables sitelinks search box in Google
- ClothingStore (LocalBusiness) — address (Dhaka), geo, openingHours, telephone
- Product — on /product/:id; price, availability, rating, offers
- BlogPosting — on /blog/:slug; datePublished, author, image, wordCount
- BreadcrumbList — on all deep pages
- FAQPage — on /faq (and any product page with FAQ accordion)

Technical SEO

- React Helmet Async: per-route <title>, <meta description>, canonical, OG, Twitter card
- Sitemap: GET /api/sitemap.xml — includes all products + blog posts with lastmod + image tags
- robots.txt: disallows /admin/, /checkout/, /cart/; allows everything else
- Bilingual keyword targeting: English + Bengali (তৈরীকৃত চম্পু পোশাক, তৈরীকৃত চম্পু পোশাক, তৈরীকৃত চম্পু পোশাক)
- Canonical URL: always https://trynexshop.com (prevents dev/staging duplicate content)
- Open Graph: image, title, type, url per route; og:locale = en_US with hreflang note

Performance

- Vite manual chunking: vendor-motion (framer-motion), vendor-query (@tanstack), vendor-radix (all Radix)
- React 19 concurrent mode: Suspense boundaries around 3D viewer and lazy-loaded routes
- Image optimisation: product images served from R2 with Cloudflare CDN caching
- Preload hints in index.html: garment mockup PNGs with fetchpriority=high for LCP
- DNS prefetch: Google Fonts, GTM, FB Connect
- PWA: service worker (Vite Plugin PWA) caches static assets, offline fallback page
- React Query: staleTime = 5 min for products, 10 min for settings; background refetch on focus
- Lazy loading: React.lazy() for all admin pages and heavy studio components (3D viewer, AI panel)

§13 Object Storage & File Handling

3 (lowest)
?

Replit Object Storage
AWS S3

No cloud keys set
AWS_ACCESS_KEY_ID set (no R2)
R2_ACCESS_KEY_ID set

Dev environment only
Alternative S3-compatible provider

- R2 bucket name from R2_BUCKET env var
- S3 endpoint: `https://{accountId}.r2.cloudflarestorage.com`
- Object keys follow pattern: `products/{id}/{filename}`
- Public CDN: if `R2_PUBLIC_BASE_URL` is set
- for public images)
- Private design files: always presigned (15-min)
- Boot guard: server hard-exits with error if NO
- sharp integration: thumbnails generated server

§14 Caching Strategy

Upstash Redis (Server-Side)

- Client: @upstash/redis REST client (HTTP, not Redis)
- In-process Map fallback: if `UPSTASH_REDIS_REST_URL` is not set
- Cached: site settings (60s TTL), product list
- Cache invalidated on every write via cache.
- Rate limit counters stored in Redis using slice

React Query (Client-Side)

- staleTime: products = 5 min, settings = 10 min
- Background refetch on window focus: enabled
- Optimistic updates: cart add/remove updates
- Query key structure: `['products', filters]`, `['products', filters, sort]`
- Suspense mode used for 3D viewer texture

§15 Email, Notifications & Integrations

Telegram Bot

- TELEGRAM_BOT_TOKEN + TELEGRAM_CHAT_ID env vars required
- Sends notifications on: new order placed, order status change, new review submitted
- Admin can test the bot via admin panel !' Settings !' Telegram test
- Webhook endpoint: POST /api/telegram-webhook for two-way messaging (future feature)

Facebook Integration

- Admin panel page: /admin/facebook-import
- Imports Facebook Page posts as draft blog posts or product listings
- Requires FACEBOOK_APP_ID and page access token configured in admin settings

Google / Social Auth

- Google OAuth: GOOGLE_CLIENT_ID required; callback at /api/auth/google
- Facebook OAuth: FACEBOOK_APP_ID required; callback at /api/auth/facebook
- Both create or link customer accounts on first login

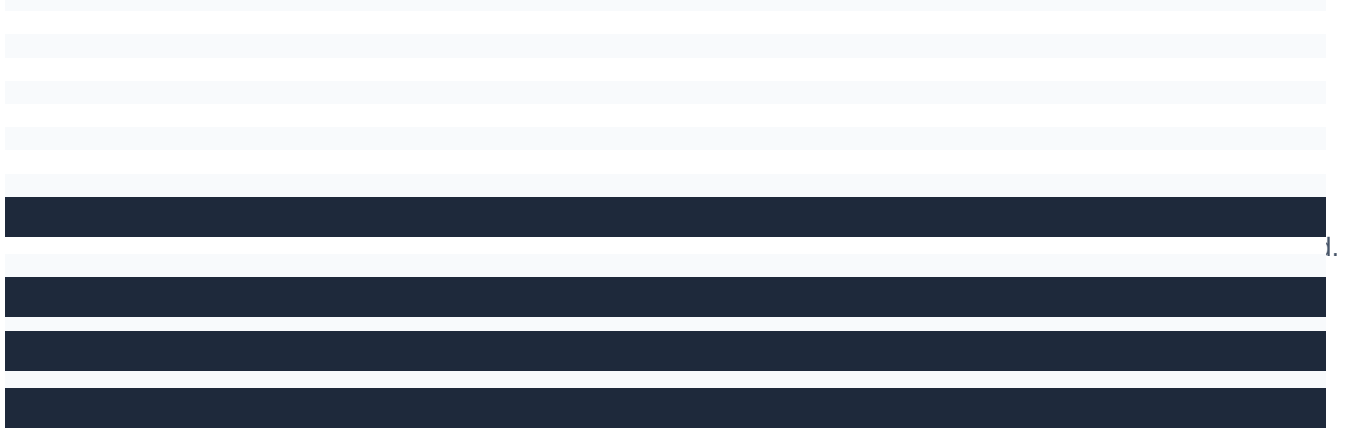
AI Tools

- Background removal: fal.ai rembg API !' WASM browser fallback (no server key needed)
- HD Upscale: fal.ai 4× super-resolution (requires FAL_API_KEY)
- AI Art: POST /api/ai/generate !' configurable OpenAI-compatible endpoint
- Admin AI Assistant: streaming SSE chat; AI can preview and execute DB operations

§16 CI/CD & Deployment Architecture

```

graph TD
    DeployAction[Deploy Action] --> PushMain[Push main (manual)]
    DeployAction --> PullRequest[Pull Request to main]
    PushMain --> TypeCheck[TypeCheck + Build (full build verification)]
    PullRequest --> SmokeTest[Smoke Test (if: always)]
    TypeCheck --> CFPages[CF Pages]
    TypeCheck --> CFWorker[CF Worker, deploy jobs (continue-on-error)]
    CFPages --> Project[trynex-lifestyle-shop project]
    CFWorker --> RenderPlus[Render+]
    SmokeTest --> RenderPlus
    Project --> RenderPlus
    RenderPlus --> DeploySequence[Deploy Sequence (deploy.yml)]
    DeploySequence --> RequiredGitHub[Required GitHub]
  
```

§18 GitHub Actions — Common Failures & Fixes

Smoke test fails on healthz
Render build state 00:00:00
Health check uses & h_not exit 1 — non-blocking
Updated: deploy skipped with warning if secret absent

Setting Up GitHub Secrets

- Go to: [https://github.com/\[owner\]/\[repo\]/settings/secrets](https://github.com/[owner]/[repo]/settings/secrets)
- Add New Repository Secret for each required variable.
- CLOUDFLARE_API_TOKEN: CF Dashboard !' My Profile Account:Read)
- CLOUDFLARE_ACCOUNT_ID: CF Dashboard !' bottom
- RENDER_DEPLOY_HOOK_URL: Render Dashboard !
- All other vars: copy from Replit Secrets panel
- Once set, re-run the last failed workflow: Actions !' Select

§19 Admin Onboarding Checklist

Initial Setup

- & Set ADMIN_PASSWORD in Replit Secrets (and Render env vars for production)
- & Set JWT_SECRET and ADMIN_JWT_SECRET (32+ chars) in Replit Secrets
- & Configure Cloudflare R2 bucket; set R2_* env vars
- & Set ALLOWED_ORIGINS to https://trynexshop.com in production
- & Set DATABASE_URL to production Neon or Render PG connection string
- & Add GitHub Secrets (see §18) to enable CI/CD
- & Verify custom domain: CF Dashboard !' DNS !' point trynexshop.com to CF Pages URL

First-Time Admin Setup

- & Login at /admin/login with ADMIN_PASSWORD
- & Enable TOTP 2FA (Admin !' Settings !' Security !' Enable 2FA) — scan QR in Google Authenticator
- & Set Studio Prices in admin Settings (T-Shirt Price, Mug Price, Bottle Price)
- & Set Studio Colours list (comma-separated hex values for Design Studio swatches)
- & Upload hero banner images, set CTA text and links
- & Seed initial categories (T-Shirts, Hoodies, Mugs, Caps, Accessories, Gift Hampers)
- & Add first products with images (upload to R2 via admin Products panel)
- & Write first blog post (admin Blog !' New Post !' use AI Assist for first draft)
- & Add Testimonials (admin Testimonials !' Create)
- & Configure Telegram notification bot (Settings !' Telegram !' Test)

SEO Setup

- & Register https://trynexshop.com in Google Search Console
- & Submit /api/sitemap.xml URL to Google Search Console
- & Register in Bing Webmaster Tools; submit sitemap
- & Add Google Analytics ID in admin Settings !' Tracking
- & Add Meta Pixel ID in admin Settings !' Tracking
- & Verify OG image (1200×630px) loads at https://trynexshop.com/og-image.jpg
- & Check structured data with https://search.google.com/test/rich-results

§20 Replit Independence

Secret management
Cache

Replit Secrets panel
In-process Map
Replit Object Storage sidebar

Render env vars / GitHub Secrets
Upstash Redis, REST
Cloudflare R2 (hard-exits if not set in prod)

Production Boot Guards

- Hard-exit if storage backend resolves to 'replit' in NODE_ENV
- Hard-exit if ALLOWED_ORIGINS is not set in production
- Hard-exit if ADMIN_JWT_SECRET is < 32 chars in prod
- Hard-exit if any of DATABASE_URL, JWT_SECRET, ADMIN_JWT_SECRET is not set
- Warns (no exit) if GOOGLE_CLIENT_ID, FACEBOOK_CLIENT_ID are not set

Migration to New Host

1. Export database: `pg_dump $DATABASE_URL > try`
2. Set up new PG instance & set DATABASE_URL to new host
3. First server start will apply all 12 Drizzle migrations
4. Configure R2_* vars (or change to S3 / another S3-compatible)
5. Set all required env vars (see §17) — server will log
6. Run seed script: `node scripts/seed.mjs` (optional —)
7. Update ALLOWED_ORIGINS and domain in Cloudflare

